

Jacob Jerry Arackal

Thodupuzha, Kerala, India

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Highly motivated Full Stack Engineer with 1+ year of experience developing scalable web and AI-driven applications. Expertise in Next.js, Python, and modern MERN stack technologies, coupled with a strong foundation in AI/ML frameworks like Langchain. Proven ability to optimize system performance, solve complex technical challenges, and drive significant user engagement in Agile environments.

Work Experience

Techfriar Technologies

♥ Kochi

Junior Software Developer

📅 May 2024 – Mar 2025

- Successfully developed **responsive web applications** using Next.js, CSS, Express, and MongoDB, leading to a 15% boost in user engagement.
- Integrated **JWT-based authentication** for secure, role-based multi-user access.
- Reduced **MongoDB** query response time by 20% through query restructuring and indexing.
- Collaborated on **RESTful API** development with Node.js and Express.js, ensuring seamless data flow between frontend and backend components.
- Led Agile ceremonies and conducted **code reviews** to improve team collaboration and enhance code quality.

Microsoft

♥ Remote

Future Ready Talent Intern

📅 Jan 2022 – June 2022

- Leveraged **Azure Technologies** to create actionable solutions for specific business challenges, resulting in improved operational efficiency.
- Built a machine learning model on Microsoft Azure using Python, NumPy, and Pandas to predict automobile prices.

Education

Amal Jyothi College of Engineering

♥ Kottayam

Bachelor of Technology - **Information Technology**

📅 July 2019 – July 2023

Projects

ProCoder | Python, Langchain, Ollama, Code Llama, Gradio

📅 Dec 2023 – Feb 2024

- Designed a real-time coding assistant using **RAG** architecture with Code Llama for **contextual code generation** and suggestions.
- Integrated Langchain and Ollama for conversational flow, enabling debugging and syntax analysis.
- Deployed an interactive UI using Gradio, supporting **live feedback for code**.

Super AI Document | Python, Streamlit, Langchain, Google Gemini Pro -API, Faiss

📅 July 2023 – Sep 2024

- Developed a **Retrieval-Augmented Generation (RAG)** application for real-time PDF analysis and contextual insight extraction.
- Leveraged Google Gemini Pro API and Langchain for advanced **question-answering over document content**.
- Optimized retrieval accuracy by implementing **FAISS** for efficient document chunking and **vector similarity search**, demonstrating strong algorithmic problem-solving.

GR Website | Python, Flask, HTML, CSS, JavaScript, Bootstrap, SQL

📅 Jan 2023 – June 2023

- Developed a **full-stack web application** using Flask, serving a REST API with HTML, CSS, and JavaScript for the frontend.
- Designed an intuitive **UI/UX** with HTML/CSS and Bootstrap, implementing JavaScript for dynamic user interactions.
- Enabled calculation of the **Golden Ratio** using custom image uploads.

Skills

Programming: Python, JavaScript, HTML, CSS, SQL, MongoDB, Node.js, GraphQL, PostgreSQL

Frameworks & Libraries: Next.js, React, Express, Langchain, Flask, Bootstrap.

Architecture & Design: Object-Oriented Programming, Clean Architecture, UI/UX Optimization, Ant-D

Tools & Technologies: Git, GitHub, JIRA, Anaconda, Docker, Minio, AWS S3, Google Cloud, FAISS

Soft Skills: Problem Solving, Time Management, **Multitasking**, Leadership, Work Ethics, Strong Debugging & Log Analysis